Employee Registry System

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to describe the functional and structural requirements of the Employee Registry System. This system is designed to manage employee information including personal details, spouse information, and children records. It provides a centralized interface for employers or administrators to create, update, search, and export employee data efficiently.
This document outlines the system scope, entity relationships, edge cases handled by the application, and assumptions made during development.

1.2 Intended Audience

This document is intended for:

- Software developers maintaining or extending the system
- Project reviewers evaluating the application
- System administrators responsible for deployment
- Stakeholders interested in system capabilities

1.3 System Overview

The Employee Registry System is a web-based application consisting of:

- **Backend:** ASP.NET Core Web API
- **Frontend:** React (Vite)
- **Database:** PostgreSQL
- **PDF Generation:** QuestPDF

The system allows users to manage employees and their family information through a structured interface while enforcing validation rules and ensuring data consistency.

2. System Scope

2.1 What the System Does

The system provides the following capabilities:

Employee Management

The system allows administrators to:

- Create new employees
- Update existing employee information
- Delete employee records
- Search employees by name, department, or NID
- View employee family information
- Automatically seed initial employee records into the database

Each employee record includes:

- Name

- National ID (NID)
- Phone number
- Department
- Basic salary

Family Management

The system allows the addition and management of family members for each employee.

Spouse

Each employee may have only one spouse. The spouse record contains:

- Name
- NID

Children

Each employee may have multiple children, and each child record contains:

- Name
- Date of birth

Search and Filtering

The system includes a debounced search functionality that allows users to filter employees dynamically based on:

- Employee name
- NID
- Department

Search results update automatically after a short delay to improve performance and user experience.

PDF Export

The system supports PDF generation for:

Employee List

- Export the currently filtered employee table to a PDF document.

Individual Employee CV

- Generate a PDF document containing detailed employee information including:
    - Personal information
    - Spouse information
    - Children details

Role-based Interface

The system supports two interface modes:

Admin Mode

- Full CRUD access
- Can add employees
- Can manage family records
- Can update employee information
- Can delete employees

Viewer Mode

- Read-only access
- Cannot modify data

2.2 What the System Does NOT Do

The system intentionally does not include the following features:

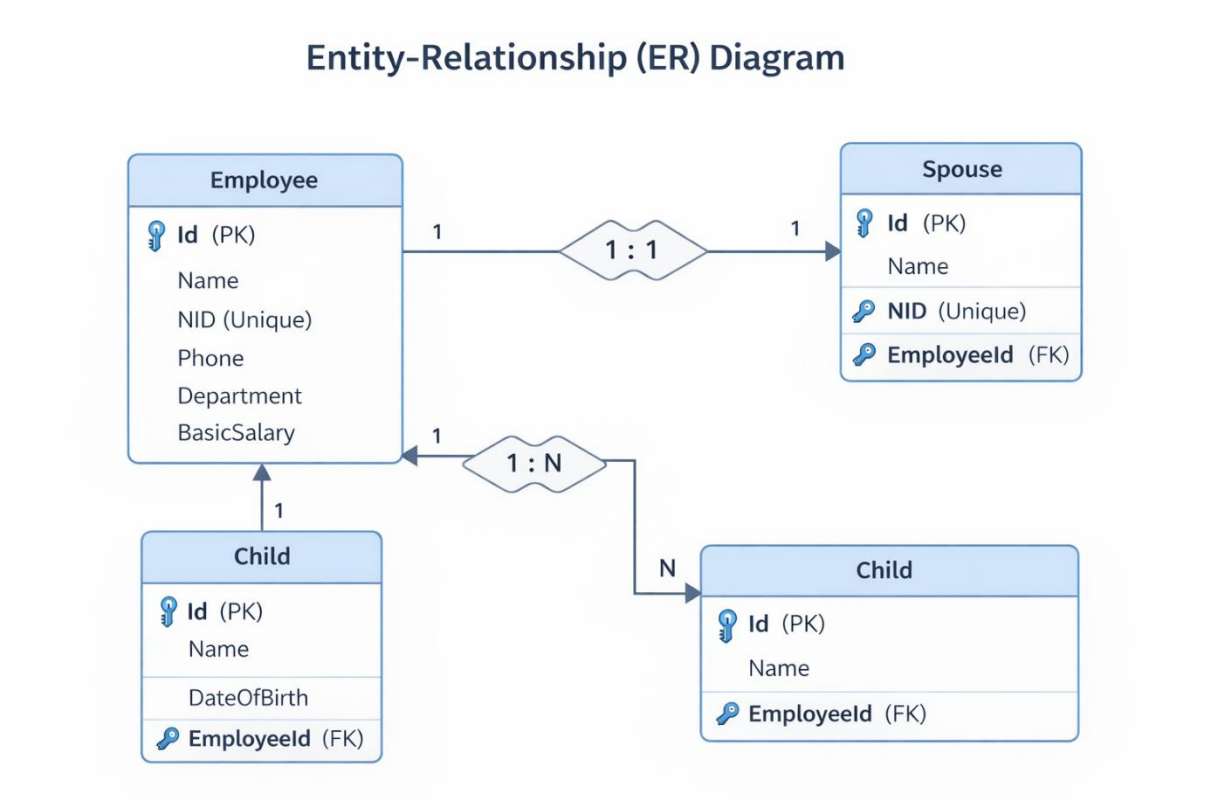
- User authentication and login system
- Payroll calculations or tax management
- File uploads or employee photos
- Multi-company or multi-organization support
- Advanced analytics or reporting dashboards

These features are outside the scope of the current implementation.

3. Entity-Relationship Diagram

The system uses three primary entities:

- Employee
- Spouse
- Child



Relationship Explanation

Employee → Spouse (One-to-One)
Each employee can have only one spouse.
The spouse entity references the employee through EmployeeId.
Employee → Child (One-to-Many)
Each employee can have multiple children, but each child belongs to exactly one employee.

4. Edge Cases

The system includes validation mechanisms to handle several potential edge cases.

Duplicate NID

The system prevents duplicate NID values.
Cases handled:

- Creating an employee with an existing NID
- Adding a spouse with an NID already used by an employee
- Adding a spouse with an NID already used by another spouse

If duplication occurs, the system returns an error message: NID must be unique

Invalid Phone Numbers

Employee phone numbers must follow the Bangladeshi phone number format.
Invalid phone formats are rejected by the system validation

Missing Required Fields

The system validates required inputs such as:

- Employee name
- Department
- Spouse name
- Spouse NID
- Child name
- Child date of birth

If required fields are missing, the user receives a clear validation message.

Multiple Spouse Prevention

Each employee is allowed only one spouse.
If a second spouse is attempted to be added, the system rejects the request.

Child Data Validation

Children records require both:

- Name
- Date of birth

If either field is missing, the system prevents creation.

Invalid Salary Input

If the salary field is left empty during employee creation, the system automatically assigns a default value of 0.

Search Edge Cases

The search system handles:

- Partial text matches
- Case-insensitive searches
- Empty search input (returns all employees)

5. Assumptions

Several assumptions were made during system design and implementation.

Single Organization Usage
The system assumes that it is used within a single organization. Multi-company support is not implemented.

One Spouse per Employee
An employee is assumed to have only one spouse within the system.

NID Uniqueness
All NID values must be unique across employees and spouses.

Children Records
Children records are stored only for identification purposes and do not include additional details such as school or guardianship.

Role Selection
The Admin/Viewer role toggle is implemented only on the frontend and does not include authentication or authorization logic.

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Seed Data
The system assumes the database may initially be empty. To support testing and demonstration, the application seeds 10 employee records automatically when the application starts.

6. Technology Stack
The system is built using the following technologies:

Layer	Technology
Backend	ASP.NET Core Web API
Frontend	React (Vite)
Database	PostgreSQL
PDF Generation	QuestPDF
ORM	Entity Framework Core

7. Conclusion
The Employee Registry System provides a simple yet structured platform for managing employee and family data. It ensures data integrity through validation rules and supports efficient data access through search and filtering capabilities. The system architecture separates frontend and backend responsibilities, allowing scalability and maintainability. Future improvements could include authentication, advanced reporting, and enhanced family relationship management.